Rack-Level Micro-Climate & Growth Optimizer for Oyster Mushroom Farms

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Project Proposal Report

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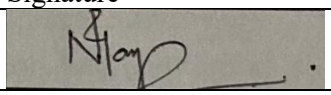
March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Oyster mushroom farming is becoming increasingly popular in Sri Lanka because it requires relatively low investment and can provide a steady source of income for small and medium-scale farmers. However, mushroom farms are highly sensitive to fungal diseases that can spread quickly across cultivation racks. At present, most farmers identify these diseases through manual visual inspection. This process is slow. Because of this delay in detection, infections can spread rapidly from one mushroom bag to multiple racks. This leads to losses, contamination of healthy mushroom bags, increased operational costs, and reduced product quality. Although several smart farming systems have been developed, most existing systems are not specifically designed to detect the major infections commonly found in Sri Lankan oyster mushroom farms. Many systems also do not support rack-level contamination tracking. This research proposes an image-based disease detection system for oyster mushroom farms. In this component, farmers will capture images of mushroom bags using a mobile application, and the system will classify mushroom bags as infected with green mold or black mold. In addition, the system will allow farmers to mark infected mushroom bags and track contamination on a rack-level basis, making it easier to monitor the spread of diseases within cultivation racks. The system will use a YOLOv8 deep learning model implemented in Python for image-based disease detection. React Native will be used for image capture and result visualization, while FastAPI will handle backend communication with the disease detection model. MongoDB will be used as the database to store rack information, and contamination records. Early disease detection can help reduce crop losses and improve the quality of mushroom production, which can increase farmers' income and farm productivity. The proposed system can also support the adoption of smart farming technologies in Sri Lanka.

Keywords: Oyster Mushroom Farming, Disease Detection, YOLOv8, Rack-Level Monitoring, Smart Agriculture

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Table of Contents

DECLARATION 2

ABSTRACT 3

ACKNOWLEDGEMENT 4

LIST OF TABLES 7

LIST OF FIGURES 8

LIST OF ABBREVIATIONS 9

INTRODUCTION 10

1.1 Background and Context 10

1.2 Literature Review 12

1.3 Research Gap 13

OBJECTIVE 15

2.1 Main Objective 15

2.2 Specific Objectives 15

METHODOLOGY 16

3.1 Research Approach 16

3.2 Technical Methods and Algorithms 16

3.3 Tools and Platforms with Justification 17

3.4 Data Collection and Ethical Considerations 17

3.5 Validation Strategy and Performance Metrics 18

HIGH-LEVEL SYSTEM ARCHITECTURE 19

4.1 Overview of Proposed Solution 19

4.2 System Diagram of Components 19

4.3 Data Flow Explanation 20

4.4 Scalability and Security 21

USER REQUIREMENTS 22

5.1 User and Stakeholder Identification 22

5.2 Functional Requirements 22

5.3 Non-functional Requirements 22

COMMERCIALIZATION PLAN 24

6.1 Target Market and Customer Persona 24

6.2 Value Persona 24

6.3 Revenue Model 24

6.4 Pricing Strategy 25

6.5 Competitive Advantage 25
6.6 Intellectual Property Considerations 26
BUDGET AND JUSTIFICATION 27
WORK BREAKDOWN STRUCTURE (WBS) 29
GANTT CHART 30
REFERENCES LIST 31
APPENDICES 33

LIST OF TABLES

Table 1 Estimated Budget 28
Table 2 Risk Analysis Table 38

LIST OF FIGURES

Figure 1 High-level system architecture 20
Figure 2 WBS of the Complete Project 29
Figure 3 Gantt Chart 30
Figure 4 Observation Checklist 33
Figure 5 Sequence diagram 34
Figure 6 Capture Image UI Mockup 35
Figure 7 History & Reports UI Mockup 36
Figure 8 Rack Tracking UI Mockup 37

LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
CNN	Convolutional Neural Network
CO_2	Carbon Dioxide
ESP32	Espressif 32-bit Microcontroller
FastAPI	Fast Application Programming Interface
GLCM	Gray Level Co-occurrence Matrix
IoT	Internet of Things
LKR	Sri Lankan Rupee
LSTM	Long Short-Term Memory
MongoDB	Mongo Database
NoSQL	Not Only SQL
SDG	Sustainable Development Goal
SaaS	Software as a Service
TIM	Technology and Information Management
UI	User Interface
YOLOv5	You Only Look Once Version 5
YOLOv8	You Only Look Once Version 8

INTRODUCTION

1.1 Background and Context

Oyster mushroom farming has become an increasingly important agricultural activity in Sri Lanka due to its low initial investment, short cultivation cycle, and potential to generate a steady income for small and medium-scale farmers. However, oyster mushroom cultivation is highly sensitive to fungal infections, making disease management one of the most critical challenges faced by farmers.

At present, most farmers identify diseases through manual visual inspection. This process is slow and depends heavily on the farmer's experience. Because of this delay in detection, infections can spread rapidly from one mushroom bag to multiple racks. This results in crop losses, contamination of healthy mushroom bags, increased operational costs, and reduced product quality. In severe cases, entire cultivation batches may need to be removed, directly affecting farmers' income and overall farm productivity. Therefore, the problem is important not only to mushroom farmers but also to farm managers, agricultural support institutions, and the wider agricultural sector, as it affects productivity, income generation, and efficient food production.

Although several smart farming systems have been developed, many existing solutions mainly focus on monitoring environmental conditions such as temperature and humidity. They are not specifically designed to detect the major fungal infections commonly found in Sri Lankan oyster mushroom farms. In addition, many systems do not support rack-level contamination tracking, even though disease spread in mushroom farms often occurs within specific racks.

Oyster mushroom farming is important both globally and locally, which highlights the need for better disease management in cultivation. At the global level, the mushroom market was reported at 19.65 million tonnes in 2025 and is projected to grow to 37.58 million tons by 2034, showing the increasing economic importance of mushroom production worldwide [3]. In Sri Lanka, mushroom cultivation also plays a meaningful role in small-scale agriculture. A local study reports that 87% of farmers' mushroom fields were affected by pests and diseases during the last season, while 80% of cultivations still experienced little or moderate yield loss even after taking control measures [4]. The same study also found that 96.2% of the surveyed farmers cultivated American oyster mushrooms, confirming the strong relevance of oyster mushroom farming in the local context [4].

To address the identified problem, this research proposes an image-based disease detection system for oyster mushroom farms. The proposed solution will consist of several main parts. First, farmers will use a mobile-based interface to capture images of mushroom bags inside cultivation racks. Second, the captured images will be sent to a backend service for processing. Third, a YOLOv8 deep learning model will analyze the images and detect visible signs of major fungal infections such as green mold (Trichoderma) and black mold (Aspergillus niger). Fourth, the detection results will be returned to the user so that infected bags can be identified quickly. In addition, the system will allow farmers to mark infected bags and track contamination on a rack-level basis, making it easier to observe the spread of infection and take corrective action. A database will be used to store rack information, detection results, and contamination records for monitoring purposes.

The study “IOT-based Monitoring System for Oyster Mushroom Farms in Sri Lanka” (2022) presented a smart farming solution tailored to the local oyster mushroom cultivation context. The system integrated environmental monitoring, harvest time detection, disease detection, and yield prediction using IoT devices together with CNN and LSTM models [5]. Although the study showed promising results, several important limitations remain. The disease detection component relied on earlier CNN-based image classification approaches rather than more advanced real-time object detection models such as YOLOv8, which are better suited for faster detection, stronger localization, and more practical deployment in dynamic farm environments. In addition, the system was not designed to track contamination at the rack level, making it difficult to identify exactly where infections begin and how they spread across different cultivation sections. This reduces its usefulness for targeted intervention.

Another study, “Automated Disease Detection in Oyster Mushroom Cultivation Using Deep Learning Technique” (2025) proposed a YOLOv5-based real-time image classification model to distinguish healthy and diseased mushrooms [6]. The study demonstrated strong detection performance and showed the value of deep learning for early identification of infections. However, the model only classified mushrooms into two broad categories, healthy and diseased, rather than identifying specific disease types [6].

The paper “IoT-Driven Smart Farming: Enhancing Mushroom Cultivation with Environmental Control and Image-Based Disease Detection” (2025) discussed an integrated concept using sensors, Raspberry Pi, ESP32, actuators, and image-based disease detection [7]. It reviewed several prior solutions and highlighted the benefits of combining IoT and AI for improving cultivation efficiency, reducing labour dependency, and enabling remote management [7]. However, this paper appears more conceptual and survey-oriented than an experimentally validated end-to-end implementation. Much of its contribution is based on summarizing earlier work rather than presenting a deeply tested, farm-deployed system [7]. As a result, questions remain regarding real-time reliability, deployment cost, robustness under local farm conditions, and scalability in practical mushroom farming environments.

The proposed research is aligned with Sustainable Development Goal 2: Zero Hunger, which aims to end hunger, achieve food security, improve nutrition, and promote sustainable agriculture [1]. This alignment is relevant because the proposed system uses artificial intelligence for early disease detection, which supports sustainable agricultural practices by helping farmers identify infections earlier, reduce crop losses, and improve productivity. The study also aligns with Sustainable Development Goal 9: Industry, Innovation and Infrastructure, as it applies artificial intelligence and digital technologies to support innovation in agriculture, improve productivity, and strengthen technology-driven farming practices [2]. In addition, the research supports Sustainable Development Goal 12: Responsible Consumption and Production, because the proposed AI-based system can reduce disease-related waste, improve resource use, and support more responsible production practices in oyster mushroom farming [2].

This study is also aligned with the Technology and Information Management (TIM) research cluster, as it combines artificial intelligence, computer vision, and information management technologies to solve practical agricultural problems through improved monitoring, decision-making, and digital innovation.

1.2 Literature Review

Recent studies show that researchers have increasingly applied Internet of Things (IoT), machine learning, and deep learning techniques to improve mushroom farming efficiency. Although these systems demonstrate promising results, many existing approaches still lack full integration, practical deployment readiness, and localized precision for rack-level cultivation environments.

Several scholarly studies have proposed IoT-based mushroom farming systems. Surige et al. developed an IoT-based monitoring system for oyster mushroom farms in Sri Lanka using CNN and LSTM models for environmental monitoring, disease detection, harvest time identification, and yield prediction [5]. This study is highly relevant because it addresses the Sri Lankan context. However, its disease detection approach relied on earlier CNN-based models rather than advanced real-time object detection methods such as YOLOv8. In addition, the system did not support rack-level contamination tracking, which limits its effectiveness for localized disease control and targeted intervention.

Shilpashree et al. proposed a real-time image-based disease detection system using YOLOv5 to classify oyster mushrooms as healthy or diseased [6]. This approach demonstrated the value of deep learning for rapid disease detection. However, the model only separated mushrooms into broad healthy and diseased categories instead of identifying specific infection classes, which limits detailed diagnosis.

Raghu Kumar et al. discussed an IoT-driven smart farming concept integrating environmental control and image-based disease detection [7]. The study emphasized the importance of combining sensors, Raspberry Pi, ESP32, actuators, and machine learning for smart mushroom cultivation. However, the paper is largely review-oriented and conceptual, with limited evidence of a fully validated farm-level implementation. This leaves gaps in terms of deployment reliability, rack-level monitoring, and real-time performance in practical mushroom cultivation settings.

Other studies have also contributed valuable insights. Guragain et al. introduced a low-cost centralized IoT ecosystem for enhancing oyster mushroom cultivation and reported strong disease detection performance together with yield improvement [8]. However, the system depended on high internet connectivity and was not designed specifically for rack-wise contamination tracing. Al-Talib et al. developed an IoT-based smart mushroom growing kit using ESP32 and the Blynk platform, demonstrating improvements in growth time and yield [9].

In terms of existing products and services, many smart farming solutions currently available in practice mainly provide environmental sensing, mobile alerts, and remote dashboard monitoring. Platforms built with technologies such as ESP32, Raspberry Pi, and Blynk commonly support temperature, humidity, and CO_2 monitoring, and actuator control. These systems improve convenience and visibility for farmers, but they are generally centered on basic environmental monitoring and remote observation rather than precise disease-focused decision support. Most do not provide rack-level contamination identification or localized spread tracking. Likewise, although image-based agricultural applications are increasingly used through mobile devices, many of them are designed for general crop classification or broad disease recognition and are not specifically adapted to the visual characteristics, growth stages, and contamination behaviour of oyster mushroom cultivation. As a result, there remains a gap between general-purpose smart farming tools and mobile-friendly, farm-specific solutions that support timely and localized intervention in mushroom farms.

1.3 Research Gap

Despite the progress shown in existing studies, several important elements are still missing from current smart mushroom cultivation solutions, making current approaches insufficient for practical farm use. First, there is a lack of specific disease identification in many existing detection systems. For example, Shilpashree et al. proposed a YOLOv5-based system that classified oyster mushrooms only into healthy and diseased categories rather than identifying the exact contamination type [6]. This is a significant limitation because different infections require different management responses, and a general disease alert provides only limited decision support to farmers. Evidence from the reviewed literature therefore shows that although disease presence can be detected, the level of diagnostic detail needed for practical farm intervention is still lacking.

There is also a lack of rack-level contamination tracking in current research. The Sri Lankan study by Surige et al. integrated environmental monitoring, harvest detection, disease detection, and yield prediction, but it did not support tracking contamination at the individual rack level [5]. This is important because mushroom disease spread is often highly localized and can move rapidly from one bag or rack to another.

In addition, several current studies are conceptual, survey-oriented, or prototype-level, rather than fully validated end-to-end farm deployments. For instance, Raghu Kumar et al. discussed an integrated IoT and image-based disease detection concept, but the paper mainly reviewed existing approaches and highlighted potential benefits rather than presenting a deeply tested farm-ready implementation [7]. This makes current approaches insufficient because conceptual potential does not automatically translate into dependable field performance. The literature itself shows that real-time reliability, robustness under farm conditions, cost, and scalability remain underexplored, which supports the argument that more practically deployable solutions are needed.

Another limitation is the dependence of some systems on high internet connectivity and cloud-based platforms, which may not be practical in all farming areas, especially rural settings. Guragain et al.'s low-cost IoT ecosystem and Al-Talib et al.'s smart mushroom growing kit both demonstrate the usefulness of connected monitoring platforms, but they also reflect reliance on high communication infrastructure for effective operation [8], [9]. This dependency creates a practical weakness because rural farm environments may experience unstable connectivity, interrupting monitoring continuity and response speed. The evidence from these studies suggests that connectivity-related constraints remain a real challenge for deployment.

There is also insufficient localized adaptation to the Sri Lankan farming context. These systems are often built on non-local datasets or designed for broader smart agriculture settings, which may not fully reflect Sri Lankan disease patterns, cultivation layouts, environmental variation, or farmer workflows. This provides evidence that the existing knowledge base is still only partially aligned with local requirements, creating room for a more context-specific solution.

Small datasets can restrict variation in lighting, angle, background, and disease stage, which affects how well a model generalizes outside test conditions. This is important because a model that performs well in controlled evaluation may still struggle in day-to-day farm use. The literature therefore provides clear evidence that dataset limitations remain a barrier to reliable real-world deployment.

The novelty of the proposed research lies in its attempt to move beyond general disease recognition toward a more practical and context-aware solution for oyster mushroom farming. Evidence from prior studies shows that many existing systems either provide only broad healthy/diseased classification [6], depend on earlier CNN-based approaches [5], or remain conceptual rather than operationally validated [7]. In contrast, this research emphasizes specific contamination recognition using a more advanced real-time object detection approach. By applying YOLOv8 in a mobile-assisted setting, the study introduces a more modern detection method that is better suited for speed, efficiency, and real-time usability under farm conditions. The research is also novel in its focus on the Sri Lankan oyster mushroom farming context, where local farm structures, disease patterns, and operational constraints differ from those assumed in many broader smart agriculture studies. This local relevance strengthens the originality of the work and makes the proposed solution more meaningful for practical application.

The creativity of the research is reflected in the way it combines technical and practical dimensions into one focused solution. Evidence from the reviewed literature shows that earlier systems rarely support rack-level contamination awareness [5], and many do not provide the level of specificity required for targeted farmer action [6].

OBJECTIVE

2.1 Main Objective

To design, develop, and evaluate, within the proposed research period, a mobile-assisted YOLOv8-based image detection system that identifies green mold (Trichoderma) and black mold (Aspergillus niger) in oyster mushroom cultivation bags and supports rack-level contamination tracking, with a target detection performance of at least 95% precision and recall.

2.2 Specific Objectives

- Dataset Component - To collect and annotate a Sri Lankan farm-relevant image dataset of oyster mushroom cultivation bags containing green mold (Trichoderma), black mold (Aspergillus niger), and healthy samples.
- Detection Model Component - To train and optimize a YOLOv8 object detection model to identify and classify visible signs of green mold and black mold in oyster mushroom cultivation bags, enabling more specific disease recognition than broad healthy/diseased classification.
- Mobile Image Capture and Processing Component - To develop a mobile-assisted image capture and processing workflow that allows farmers to capture mushroom bag images and receive detection results in a practical and accessible manner.
- Rack-Level Tracking Component - To design and implement a rack-level contamination tracking mechanism that records infected bag locations and supports localized monitoring of contamination spread within cultivation racks.
- System Evaluation Component - To evaluate the performance of the proposed detection system using standard object detection metrics such as precision, and recall, together with detection response time under varying image conditions.

METHODOLOGY

3.1 Research Approach

This study will adopt a Design Science Research approach supported by experimental evaluation. Design Science Research is appropriate because the purpose of the study is not only to analyse a problem, but also to design and develop a practical artefact that addresses it. In this research, the artefact is a mobile-assisted image-based disease detection system for oyster mushroom farms. The approach is suitable because the identified problem is practical and solution-oriented: farmers need a usable digital system that can help detect fungal contamination.

The study will also include an experimental component to evaluate the technical performance of the developed system. After designing and implementing the proposed solution, experiments will be carried out to test the YOLOv8 detection model using labelled mushroom bag images collected under farm-relevant conditions. This will allow the research to assess how accurately and efficiently the system performs in identifying green mold (Trichoderma) and black mold (Aspergillus niger). Therefore, the methodology combines artefact creation with empirical evaluation, making Design Science with experimental validation the most suitable overall research strategy.

3.2 Technical Methods and Algorithms

The core technical method used in this study will be deep learning-based object detection. The main algorithm selected is YOLOv8 (You Only Look Once version 8). YOLOv8 is chosen because it is a modern real-time object detection model that offers strong performance in terms of detection accuracy, processing speed, and localization capability. Unlike earlier classification-based approaches that only label the whole image, object detection allows the model to identify the contaminated region within the image. This is more appropriate for mushroom disease monitoring because the system must not only determine whether contamination exists, but also indicate where it appears on the cultivation bag.

The model will be trained to detect two contamination classes: green mold (Trichoderma) and black mold (Aspergillus niger). Healthy samples will also be included during dataset preparation to improve model discrimination between contaminated and non-contaminated conditions. Image annotation will be carried out using bounding boxes around visible infected regions so that the model can learn spatial patterns of contamination. During training, the dataset will be divided into training, validation, and testing subsets. Data augmentation techniques such as rotation, flipping, scaling, and brightness adjustment may be applied to improve robustness against farm-related image variation such as different angles, lighting conditions, and backgrounds.

For rack-level monitoring, a structured contamination logging mechanism will be designed. Each analysed image will be associated with metadata such as rack number, bag identifier, date, contamination type, and prediction result. This will allow the system to maintain a simple contamination history and support localized monitoring inside a cultivation area. Although the main intelligence of the system lies in the detection model, this tracking layer is important because it converts image predictions into actionable farm-level monitoring information.

3.3 Tools and Platforms with Justification

The model development process will be carried out using Python, as it provides strong support for deep learning, image processing, and backend integration. The YOLOv8 model will be implemented using the Ultralytics YOLO framework, which is suitable because it offers an efficient and well-supported environment for training, validation, and inference. Python is appropriate for this study because it is widely used in computer vision research, has extensive library support, and enables smooth integration between model development and application components.

For image annotation, Roboflow will be used to mark contaminated regions in the collected mushroom bag images. Roboflow is suitable because it supports bounding-box annotation formats required for object detection training and provides integrated tools for dataset organization and export in YOLO-compatible formats. For image handling and preprocessing, libraries such as OpenCV will be used. OpenCV provides efficient functions for image reading, resizing, augmentation, and format conversion, which support the preparation of images for model training.

The backend service will be developed using FastAPI, a lightweight and high-performance Python web framework. FastAPI enables the mobile-assisted interface to send captured mushroom bag images to the server and receive detection results generated by the YOLOv8 model. Using a backend API allows the system to separate model inference from the user-facing interface.

For data storage, MongoDB will be used to manage rack information, bag identifiers, contamination records, timestamps, and detection outputs. MongoDB is suitable for this system because it is a flexible NoSQL document-oriented database that can efficiently store structured and semi-structured data generated from image detection results. This flexibility supports rack-level contamination tracking and allows easy retrieval of historical monitoring data for analysis.

For the user interaction layer, a mobile-assisted interface will be used rather than deploying a fully embedded AI model directly on the mobile device. This approach is practical for the current scope because it reduces on-device computational requirements while still allowing farmers to use smartphones to capture images of mushroom cultivation bags. The mobile interface, developed using React Native, enables users to capture images and view contamination detection results. Captured images are sent to the backend for processing and prediction.

3.4 Data Collection and Ethical Considerations

The data used in this study will consist will include three categories: bags affected by green mold (Trichoderma), bags affected by black mold (Aspergillus niger), and healthy bags. Image data will be collected from an oyster mushroom cultivation facility located in Makandura, with the assistance of Agricultural Research Center in Makandura. Images will be captured using a mobile phone camera from different distances, angles, and lighting conditions in order to better represent practical farm usage scenarios.

The data collection process will be conducted in a structured manner to ensure consistency and relevance. Collected images will be reviewed and manually annotated based on visible signs of contamination. The dataset will then be divided into training, validation, and testing sets to support model training, tuning, and performance evaluation.

Ethical considerations in this research are relatively low-risk because the study focuses on agricultural image data rather than personal or medical information. However, several ethical aspects will still be considered. Permission will be obtained prior to collecting images from the mushroom cultivation facility in Makandura. The identity of farm owners, workers, and any sensitive operational details will not be disclosed without explicit consent. Collected data will be used strictly for academic research purposes.

3.5 Validation Strategy and Performance Metrics

The proposed system will be validated through both technical performance evaluation and practical functionality assessment. The technical validation will focus mainly on the YOLOv8 model's ability to detect and classify the selected fungal contaminations correctly. The dataset will be divided into separate training, validation, and testing subsets so that model performance can be measured on unseen data. This helps reduce bias and provides a clearer indication of generalization ability.

The main performance metrics will include precision, and recall. Precision will be used to measure how many detected contaminations are actually correct. Recall will measure how many true contaminations are successfully identified by the model. In addition, detection response time will be measured to assess whether the system performs fast enough for practical use in a mobile-assisted workflow. Beyond model metrics, the system will also be validated at the functional level. This includes checking whether images can be captured and submitted successfully, and whether predictions are returned correctly to the interface.

HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of Proposed Solution

The proposed solution is a mobile-assisted oyster mushroom disease detection and rack-level contamination tracking system designed for practical use in Sri Lankan mushroom farming environments. The system focuses on helping farmers identify visible fungal contamination by capturing images of mushroom cultivation bags and analyzing them using a YOLOv8-based object detection model. In addition to detection, the solution supports rack-level contamination tracking, allowing infected bag locations to be recorded and monitored within cultivation racks.

The solution consists of five main components. The first is the mobile interface, which allows the user to capture images of mushroom bags and submit them for analysis. The second is the backend service, developed using FastAPI, which manages communication between the mobile interface, the detection model, and the database. The third is the YOLOv8 detection engine, which processes submitted images and identifies visible signs of green mold (Trichoderma) and black mold (Aspergillus niger). The fourth is the MongoDB database, which stores rack information, bag identifiers, detection outputs, timestamps, and contamination history. The fifth is the admin or monitoring interface, which supports viewing records, rack status, and contamination logs for analysis and follow-up action.

4.2 System Diagram of Components

The proposed system includes the following high-level components:

- Farmer
- Mobile Interface (React Native) - The mobile interface is used by the farmer to capture mushroom bag images and submit them to the system. Its main parts include the camera access function, image preview, submission option, and result display screen. This component acts as the main user-facing entry point of the system.
- FastAPI Backend - The backend acts as the central controller of the system. Its main parts include the API endpoints, request handler, model inference connection, and database communication logic. It receives the image from the mobile interface, sends it to the detection engine, retrieves the results, and stores relevant records.
- YOLOv8 Detection Engine - This is the core intelligence component of the system. Its parts include the trained YOLOv8 model, preprocessing logic, inference process, and output formatting step. It analyzes each submitted image and detects whether visible contamination is present, while identifying the contamination class.
- MongoDB Database - The database stores structured system records. Its parts include collections for rack details, bag records, image metadata, detection outputs, timestamps, and contamination logs. This component supports later retrieval, monitoring, and history tracking.
- Admin/Monitoring Interface - This interface supports monitoring and review. Its parts may include rack view, contamination history display, record filtering, and summary visualization. It helps the farm manager or researcher observe system outputs over time and assess contamination patterns.
- Farm Manager/Researcher

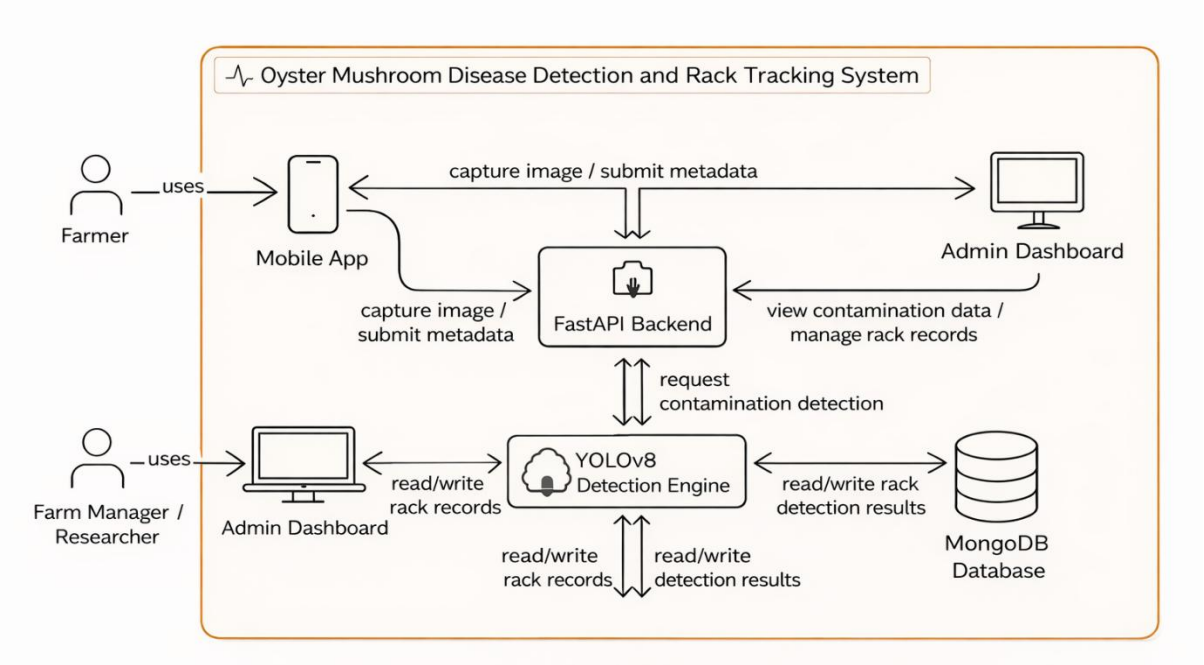


Figure 1 High-level system architecture

The components interact in a connected sequence. The farmer begins by using the mobile interface to capture an image of a mushroom cultivation bag. Once submitted, the image is sent to the FastAPI backend. The backend validates the request and passes the image to the YOLOv8 detection engine for analysis. After inference, the detection engine returns the identified contamination class and related output to the backend. The backend then stores the result in MongoDB together with rack-level information, timestamp, and record identifiers.

At the same time, the result is returned to the mobile interface so the farmer can immediately see the outcome. If contamination is detected, the record becomes part of the rack-level contamination history. Through the admin or monitoring interface, the farm manager or researcher can later access these stored records to observe which rack areas are affected, review contamination logs, and monitor spread patterns over time.

4.3 Data Flow Explanation

The data flow begins with image acquisition. A farmer captures an image of a mushroom bag through the mobile interface. This image is then packaged and transmitted to the FastAPI backend through an API request. The backend performs basic validation and sends the image to the YOLOv8 detection model.

Within the detection stage, the model processes the image and identifies visible contamination regions based on the trained classes, such as green mold and black mold. The model output is returned to the backend in the form of predicted labels and detection details. The backend then links this result with rack-related metadata, such as rack identifier, bag identifier, capture time, and detection status.

Next, this combined information is stored in MongoDB. The backend also returns the prediction result to the mobile interface for immediate user feedback. Later, the admin interface requests stored records from the backend to display contamination logs, rack-level records, and monitoring information. In this way, data flows through the system from capture, to detection, to storage, to monitoring.

4.4 Scalability and Security

From a scalability perspective, the proposed architecture is designed so that each major function is separated into logical components. This makes future expansion easier. For example, the detection model can later be improved with additional contamination classes, the database can support a larger number of records, and the system can be extended to support more farms, more racks, or multi-user access. Because the backend, model, and database are separated, the system can also be upgraded incrementally without redesigning the full solution. This modular structure supports maintainability and future research extension.

From a security perspective, the system should ensure that only authorized users can access stored contamination records and rack information. Basic authentication and controlled API access should be used for the backend and monitoring interface. Uploaded images and stored records should be handled securely, and farm-identifying information should be protected where needed. Since the system is intended for agricultural monitoring rather than sensitive personal health data, the security risk is lower than in medical systems, but responsible handling of farm data, user access, and stored records is still necessary to maintain trust and proper system use.

USER REQUIREMENTS

5.1 User and Stakeholder Identification

The proposed system involves both direct users and broader stakeholders who are connected to oyster mushroom cultivation and disease management. The primary user is the oyster mushroom farmer, who will use the mobile interface to capture images of cultivation bags, submit them for detection, and view contamination results. Another important user is the farm manager or researcher, who will use the monitoring interface to review rack-level contamination records, observe detection history, and analyze spread patterns within the cultivation environment.

Other stakeholders also have an interest in the system. These include farm owners, who are concerned with crop protection, yield stability, and operational efficiency; agricultural researchers or academic supervisors, who may use the system outputs for evaluation and analysis; and agricultural extension or support institutions, which may benefit from improved understanding of disease occurrence in oyster mushroom farms.

5.2 Functional Requirements

The functional requirements describe what the system must do to support image-based disease detection and rack-level contamination tracking.

1. The system shall allow the user to capture an image of an oyster mushroom cultivation bag using a mobile device.
2. The system shall allow the user to upload or submit the captured image for contamination analysis.
3. The system shall process the submitted image using the trained YOLOv8 detection model.
4. The system shall detect visible contamination classes supported by the model, including green mold (Trichoderma) and black mold (Aspergillus niger).
5. The system shall return the detection result to the user through the mobile interface.
6. The system shall display whether the submitted image contains a detected contamination or no contamination identified within the supported classes.
7. The system shall allow the user to provide rack identifier and bag identifier details during image submission and shall store these user-entered details together with the image timestamp and predicted contamination result.
8. The system shall maintain a contamination history log for recorded rack entries.

5.3 Non-functional Requirements

The non-functional requirements describe the quality attributes the system should satisfy.

1. Performance: The system should return image detection results within an acceptable response time for practical farm use.
2. Accuracy: The system should achieve satisfactory detection performance for the supported contamination classes when evaluated using standard object detection metrics.
3. Usability: The mobile interface should be simple and easy to use for farmers with basic smartphone familiarity.
4. Reliability: The system should operate consistently during repeated image submissions and result retrieval.
5. Security: Access to stored rack records and contamination logs should be limited to authorized users only.
6. Data Integrity: Detection records, rack details, and contamination history should be stored accurately without unintended modification or loss.

7. Maintainability: The system should be structured in a modular way so that the mobile app, backend, detection model, and database can be updated separately.
8. Scalability: The system should support future expansion, such as additional contamination classes, more farms, more racks, or increased numbers of stored records.
9. Compatibility: The mobile-assisted solution should function on commonly available smartphones used for image capture.
10. Privacy and Confidentiality: Farm-related information collected during the study should be handled responsibly and should not be disclosed without permission.
11. Robustness: The system should be able to handle some variation in image conditions, such as differences in lighting, angle, and background, within the limits of the trained dataset.

COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target market for the proposed system is small and medium-scale oyster mushroom farms in Sri Lanka, especially farms that manage multiple cultivation racks and face recurring contamination losses. The most suitable early customers are growers who already use smartphones in daily operations and want a low-cost way to detect infections earlier and monitor affected rack locations. A second potential segment is farm managers, training centers, agricultural institutes, and university-led mushroom cultivation units, which may use the system for farm monitoring, demonstrations, and disease management support.

A realistic customer persona is a Sri Lankan oyster mushroom farm owner or manager aged 25–50, operating a modest rack-based farm, using a smartphone for communication and business tasks, and looking for practical tools that reduce spoilage and avoid full-batch loss. This customer is price-sensitive, values simplicity, and is unlikely to invest in expensive industrial automation. Therefore, the proposed solution is positioned as an affordable mobile-assisted decision-support tool rather than a high-cost enterprise platform.

6.2 Value Persona

The proposed system offers value by helping farmers detect green mold and black mold earlier, identify affected bags more quickly, and maintain rack-level contamination records for more targeted intervention. This can reduce the spread of contamination, lower avoidable crop loss, and improve operational awareness inside the cultivation environment. Unlike general environmental monitoring tools, the proposed solution is more disease-focused and action-oriented, because it supports visual contamination recognition linked with rack-level monitoring.

Its value is also strengthened by local relevance. The system is tailored to oyster mushroom cultivation in Sri Lanka, rather than broad smart agriculture use cases. It is designed around a mobile-assisted workflow, which makes it more practical for real farm use where farmers already rely on phones more than laptops or specialized equipment. In commercial terms, the value proposition is: early detection, localized visibility, and lower preventable losses at a price point accessible to small farms.

6.3 Revenue Model

A realistic commercialization path is a monthly or annual subscription-based Software-as-a-Service (SaaS) model with a low onboarding fee. The subscription approach is more suitable than a one-time license because the system requires backend hosting, model inference, database storage, maintenance, and version updates.

A practical revenue structure for Sri Lanka would be:

- Starter Plan for small farms
- Standard Plan for medium farms
- Institutional Plan for training centers, research units, or multi-rack commercial farms

In addition, a small setup/training fee can be charged during onboarding to configure the farm profile, rack layout, and user guidance. Future revenue options could include dataset expansion services, additional contamination classes, or white-label deployment for agricultural institutions.

6.4 Pricing Strategy

A penetration pricing strategy is appropriate in the Sri Lankan market. Since the target users are cost-sensitive small and medium mushroom farmers, the service should be priced low enough to encourage adoption while still covering recurring costs.

Plan	Target User	Monthly Price (LKR)
Starter	Small farms, limited racks	3,500
Standard	Medium farms, more frequent usage	7,500
Institutional	Research units/larger farms	15,000

A one-time setup and training fee of LKR 10,000-20,000 can also be charged depending on farm size and onboarding effort.

This pricing is realistic because if the business serves:

- 10 Starter customers = LKR 35,000
- 5 Standard customers = LKR 37,500
- 2 Institutional customers = LKR 30,000

The total monthly revenue would be LKR 102,500, which is above the estimated monthly operating cost of LKR 75,000. That gives a modest but believable operating margin in an early commercial stage.

6.5 Competitive Advantage

The proposed solution has several practical competitive advantages. First, it is disease-focused, while many existing low-cost smart farming products mainly emphasize environmental monitoring and alerts. Second, it provides specific contamination recognition for selected fungal infections, rather than only broad healthy/diseased classification. Third, it includes rack-level contamination tracking, which improves localized decision-making and helps farmers isolate affected sections more effectively.

Another advantage is its Sri Lankan contextual relevance. Many available tools and studies are either generic, prototype-level, or based on non-local settings. In contrast, this system is designed around local oyster mushroom farming practices and mobile-based usability. That local fit can become a strong differentiator, especially if the dataset continues to grow with Sri Lankan farm images over time. The combination of mobile accessibility, localized disease focus, and rack-level record keeping creates a stronger value position than a generic smart farming dashboard alone.

6.6 Intellectual Property Considerations

The main intellectual property generated by this project may include the custom annotated image dataset, the trained YOLOv8 model weights, the mobile and backend software code, and the overall system design and workflow integration. These can potentially be protected through a combination of copyright, access control, licensing, and institutional ownership policies.

In practical terms, the source code can be protected as software copyright, while the trained model and curated dataset can be treated as proprietary research assets if they are not publicly released. Since the system builds on existing open technologies such as React Native, FastAPI, MongoDB, and YOLO-based frameworks, commercialization must also respect the license conditions of those tools and libraries.

BUDGET AND JUSTIFICATION

The proposed disease detection component requires a moderate budget covering hardware, software support, cloud/backend usage, and development effort. Since the project is designed as a research prototype rather than a fully commercial deployment at this stage, the budget focuses on essential and realistic expenses only. The cost estimation below is intended to support data collection, model development, backend integration, mobile-assisted testing, and evaluation under farm-relevant conditions.

Item	**Quantity**	**Estimated Unit Cost (LKR)**	**Estimated Total (LKR)**	**Justification**
Smartphone for image capture	1	55,000	55,000	Needed to capture mushroom bag images in realistic farm conditions
Mid-range laptop	1	90,000	90,000	Used for model training, annotation handling, preprocessing, and backend/model testing
Travel for farm data collection	4 visits	3,000	12,000	Required to collect real farm images under practical conditions
Portable storage	1	8,000	8,000	Needed for secure backup of collected image datasets, annotations, and experiment files
Roboflow annotation plan	1	0-15,000	15,000	Annotation can begin with free tier, but allocation is included in case higher annotation
Development tools (VS Code, Python, FastAPI, React Native)	-	0	0	These tools are open-source
MongoDB Community Edition	-	0	0	Free version is sufficient
Cloud storage/model hosting/API testing	1	10,000	10,000	Supports backend deployment, test hosting, and storage
Internet and mobile data	10 months	2,500	25,000	Required for syncing development work, uploading images, and testing APIs

Dataset collection and annotation effort	1 component effort block	45,000	45,000	Covers time spent collecting, reviewing, labeling, and organizing farm-relevant image data
Model development and testing effort	1 component effort block	75,000	75,000	Covers preprocessing, YOLOv8 training, tuning, validation, and performance evaluation
Mobile-backend integration effort	1 component effort block	60,000	60,000	Covers FastAPI integration, MongoDB handling, and React Native testing support

Table 1 Estimated Budget

WORK BREAKDOWN STRUCTURE (WBS)

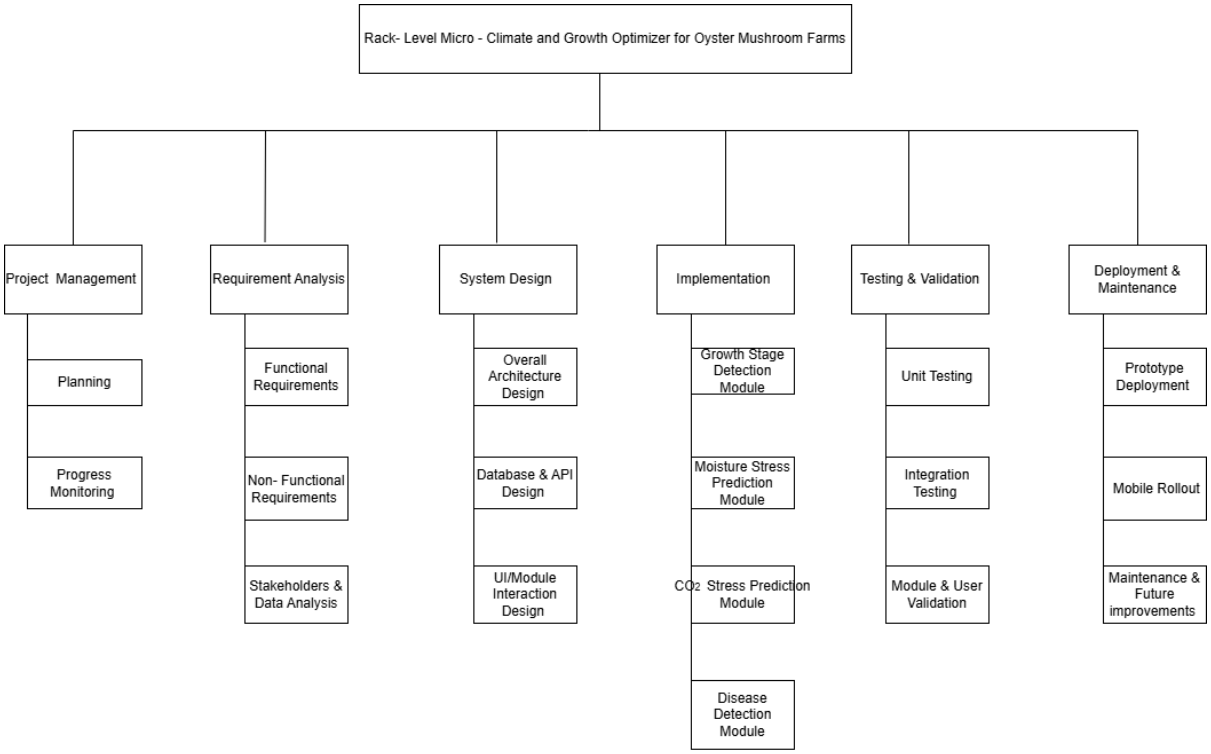


Figure 2 WBS of the Complete Project

The work is divided according to the four main components of the overall system so that each member is responsible for a clearly defined research and implementation area. Member 1 handles biological growth stage detection using image analysis. Member 2 handles rack-level moisture stress regulation using temperature-humidity interaction and rule-based control. Member 3 handles CO_2-based physiological stress prediction and image-based validation. Member 4 handles image-based disease detection and rack-level contamination tracking. In addition to component-specific work, all members contribute to common tasks such as literature review, requirement analysis, integration, report writing, presentation preparation, and final testing.

The project timeline is structured to match the major phases of the research and development process. The early phase focuses on literature review, requirements gathering, data collection planning, and initial system design. The middle phase covers data collection, annotation, model development, backend and mobile integration, and component-level implementation. The later phase is dedicated to testing, validation, system integration, refinement, documentation, and deployment preparation. This timeline is also aligned with the expected academic milestones, including 50% prototype completion by early May, full integration by August, and deployment readiness by October.

GANTT CHART

The following Gantt chart is provided to demonstrate timeline alignment of project tasks and milestone-based progress across the research period.

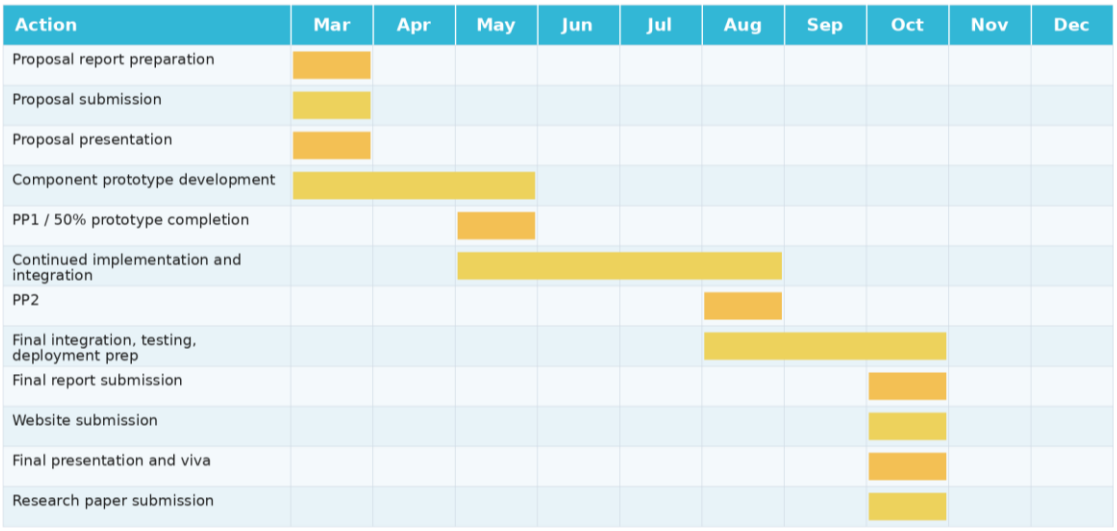


Figure 3 Gantt Chart

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APPENDICES

The following observation checklist was prepared to guide data collection during farm visits and image capture activities.

Observation Checklist

Farm Name: _____

Location: _____ Date: _____

Area to Observe	Yes	No	Comments
Environment			
Temperature within optimal range (24–28°C)	☐	☐	
Humidity level maintained (80–95%)	☐	☐	
Ventilation adequate	☐	☐	
Lighting schedule followed as planned	☐	☐	
Growing Conditions			
Moisture level of substrate maintained	☐	☐	
Mushroom racks properly organized	☐	☐	
Fresh air exchange system functional	☐	☐	
Crop Health			
Healthy mushroom growth observed	☐	☐	
Discoloration/deformities	☐	☐	
Presence of pests or insects	☐	☐	
Contamination Checks			
Signs of mold or contamination in bags	☐	☐	
Hygiene practices observed by workers	☐	☐	
Operational Practices			
Monitoring equipment functional	☐	☐	
Record-keeping updated regularly	☐	☐	

Figure 4 Observation Checklist

The following figure illustrates the supporting artefacts developed for the image-based disease detection component.

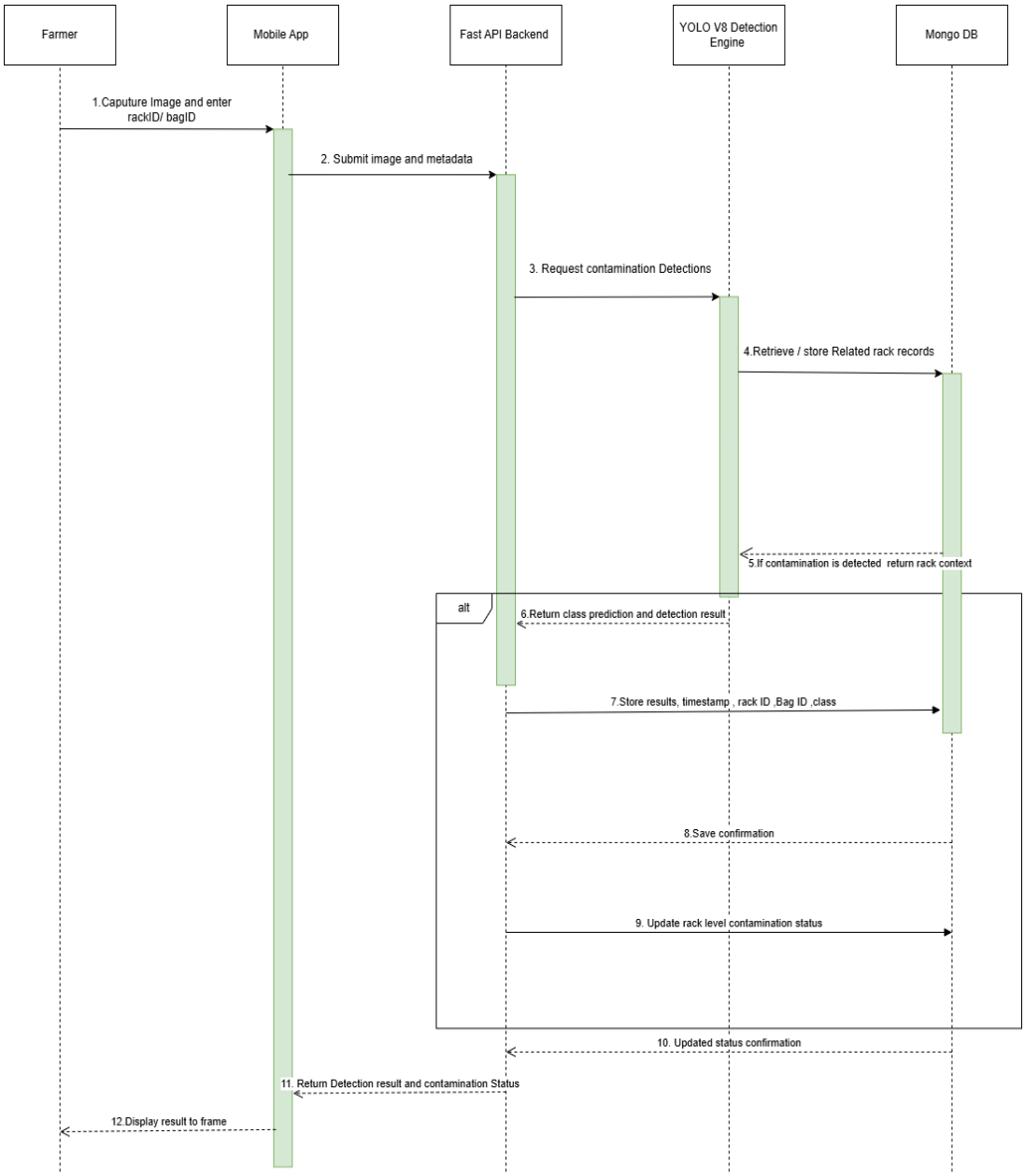


Figure 5 Sequence diagram

The following UI mockups present the preliminary mobile interface designs for key functions of the proposed system.

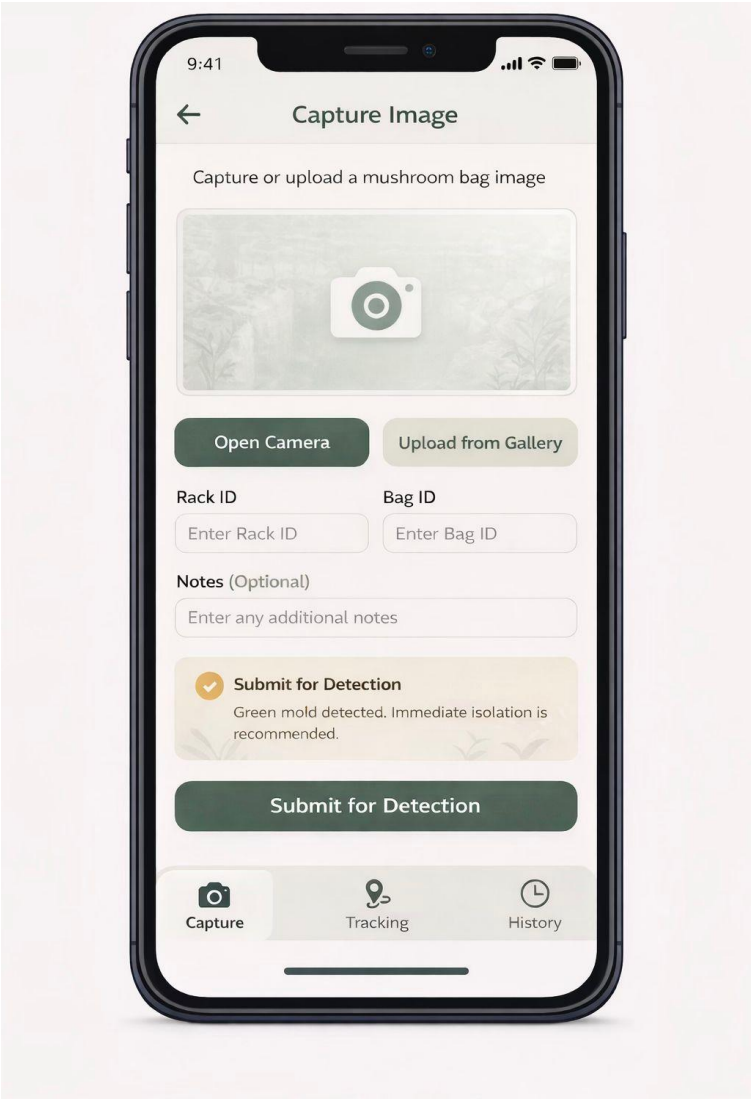


Figure 6 Capture Image UI Mockup

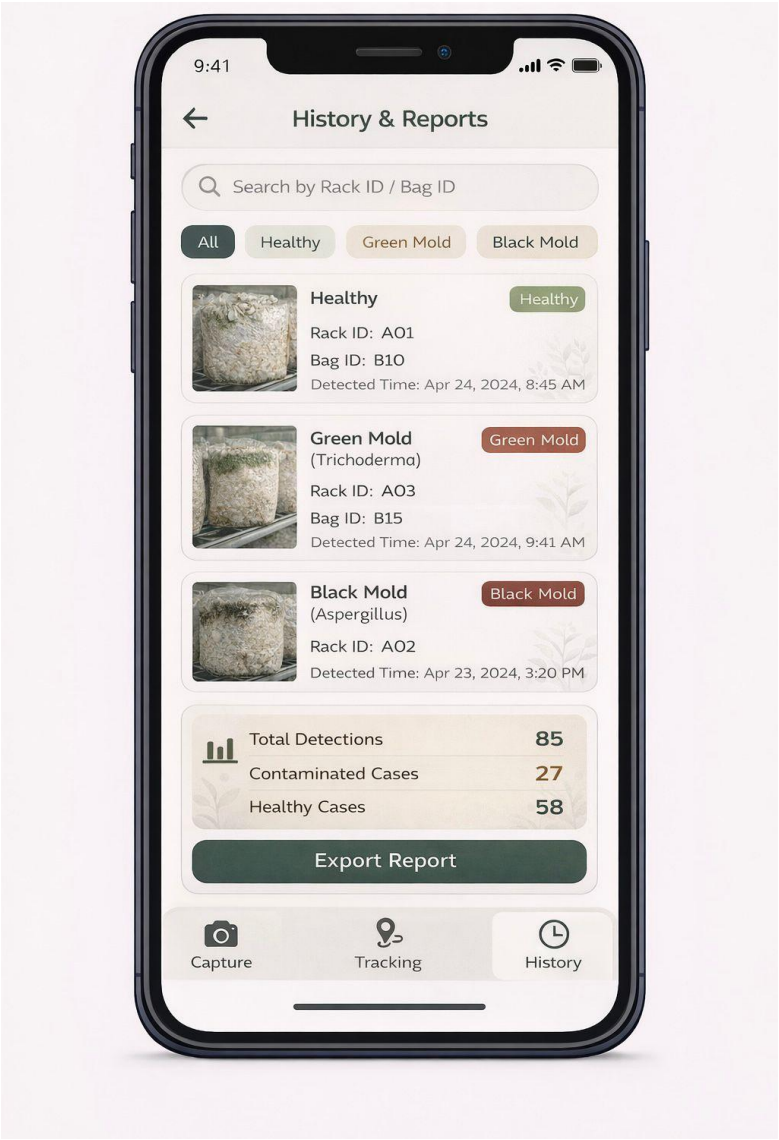


Figure 7 History & Reports UI Mockup

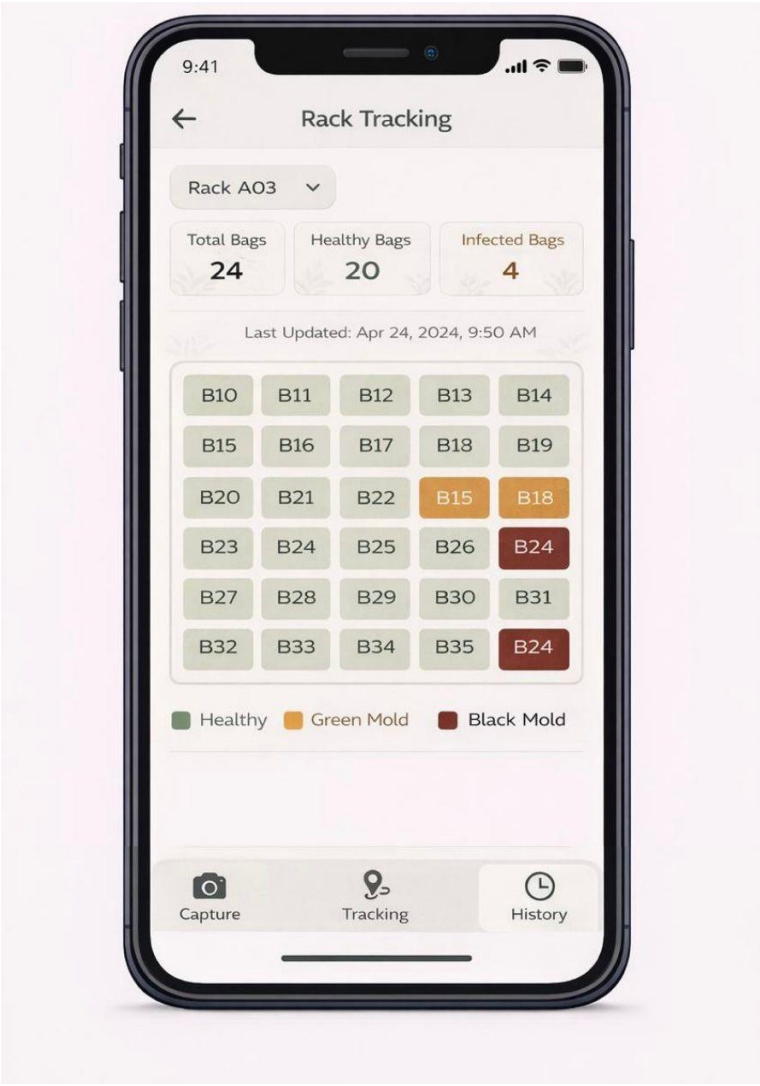


Figure 8 Rack Tracking UI Mockup

The following table summarizes key risks identified for the proposed component together with mitigation strategies planned to reduce their impact.

Risk Description	Mitigation Strategy
Similar visual appearance between contamination types causes misclassification	Increase dataset diversity, improve annotation quality, and tune the YOLOv8 model through repeated training and validation
Poor image quality due to lighting, blur, or angle variations	Collect images under varied realistic conditions and apply preprocessing and augmentation during model development
Limited time may affect full integration and testing of all system components	Follow a milestone-based work plan, prioritize core features first, and perform early prototype integration
Ethical or permission-related issues when collecting farm data	Obtain permission before data collection, avoid disclosing farm-sensitive details, and use data only for academic purposes
Hardware or software failures during development and testing	Maintain backups of source code, dataset, annotations, and documentation using GitHub and cloud/local storage

Table 2 Risk Analysis Table